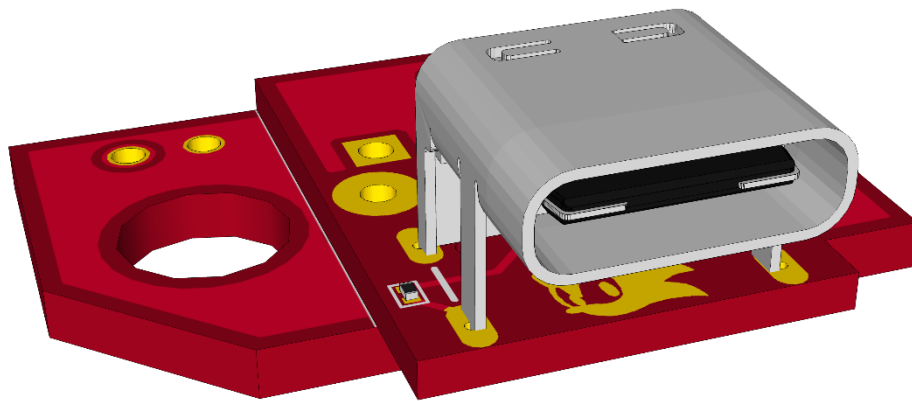


USB-C KIT FOR
MASTER SYSTEM 1
MASTER SYSTEM 2
MEGA DRIVE 1
MEGA DRIVE 2



PRODUCT

[HTTPS://SHOP.GILTESA.COM/PRODUCT/SEGA-MASTER-SYSTEM-MEGA-DRIVE-USB-C-KIT](https://shop.giltesa.com/product/sega-master-system-mega-drive-usb-c-kit)

**PLEASE READ THROUGH THESE INSTRUCTIONS
ENTIRELY BEFORE ATTEMPTING TO INSTALL**

**WARNING: IF YOU ARE NOT COMFORTABLE WITH
SOLDERING, OR PERFORMING ANY STEP IN THIS
GUIDE, DO NOT PERFORM THE INSTALL YOURSELF.
FIND SOMEONE WHO IS COMFORTABLE TO DO IT FOR
YOU.**

INDEX

DESCRIPTION	4
FEATURES	4
INCLUDED	5
RECOMMENDED / REQUIRED [NOT INCLUDED]	5
BOARD DETAILS.....	6
TEST THE BOARD!.....	7
BOARD COMPATIBILITY	8
INSTALLATION STEPS	9
PRE INSTALLATION STEPS.....	9
1. DISASSEMBLY THE CONSOLE	9
2. REMOVE UNNECESSARY PARTS.....	12
INSTALLATION STEPS.....	21
1. BOARD ASSEMBLY	21
2. DONE!	28
FREQUENTLY ASKED QUESTIONS - FAQ.....	29

DESCRIPTION

The SEGA **Master System & Mega Drive USB-C Kit** is a custom board that lets you upgrade your console by replacing the original power connector with a modern, universal USB-C connector.

Replace your old or damaged **power connector** and upgrade to modern convenience. This **USB-C mod kit lets you power your console using any standard USB-C charger**, including those for your smartphone or laptop.

This board is compatible with the following models:

- Master System 1 ⁽¹⁾
- Master System 2 ⁽¹⁾
- Mega Drive 1 ⁽¹⁾
- Mega Drive 2 ⁽¹⁾

⁽¹⁾ There were so many mainboard revisions of each model; before purchasing, make sure your console, motherboard, and connector match the ones shown in the installation guide.

FEATURES

- External power through USB-C.
- Fully compatible with any USB-C charger:
 - USB power banks
 - USB-A chargers
 - USB-C chargers
 - USB-C PD chargers
 - USB-A to USB-C cables
 - USB-C to USB-C cables

INCLUDED

- 1 USB-C board
- 1 Mega Drive 1 board adapter
- 1 Plastic cap for the USB-C board
- 5 Single-pin headers
- 4 wires

RECOMMENDED / REQUIRED [NOT INCLUDED]

- Phillips screwdriver
- Soldering iron
- Kapton tape
- Soldering tin
- Flux
- Desoldering pump
- Desoldering braid
- Isopropyl alcohol

BOARD DETAILS

This small board has many pads, but not all of them need to be soldered, as the required ones depend on the console in which the kit is installed.

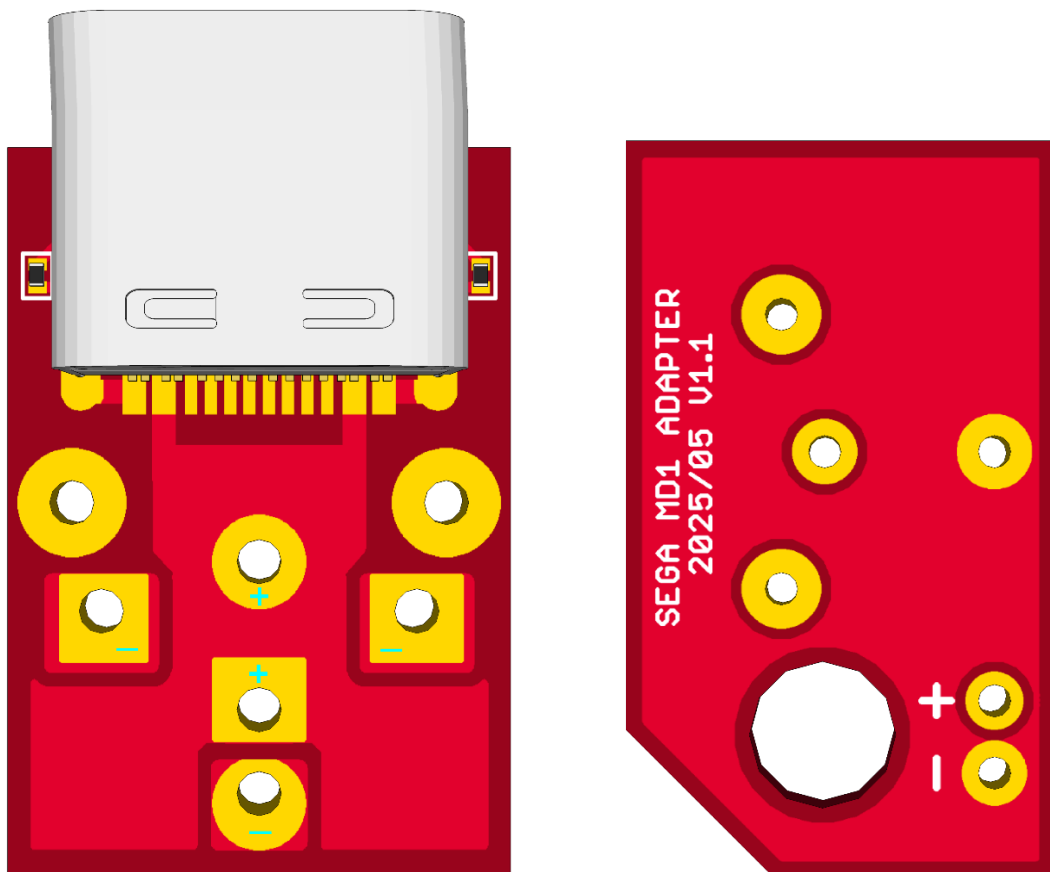
The pads with a **round shape** are for:

- Master System 1
- Master System 2
- Mega Drive 1

While the pads with a **square shape** are only for:

- Mega Drive 2

On the image, you can see marked which pads are positive (+) and which are negative (-). Although this is merely informative, during installation you only need to solder them, and that's it.

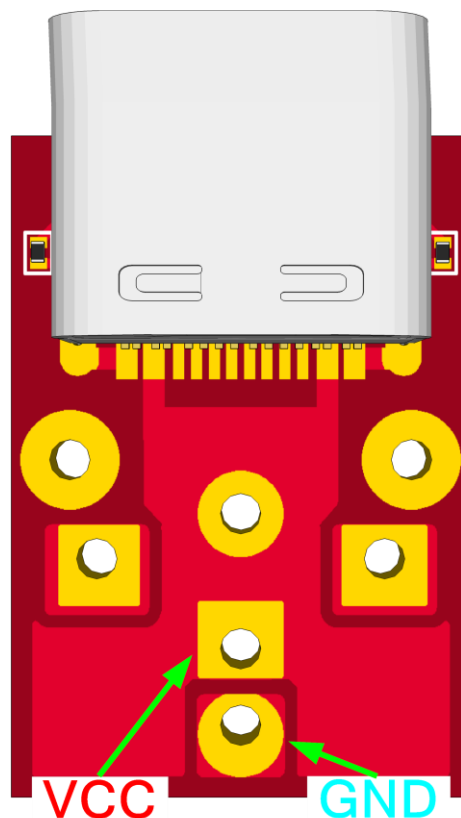


The second board included in the kit is only needed for Mega Drive 1 (on the right)

TEST THE BOARD!

Before starting the installation, you should test the board. If it doesn't work contact me [for a replacement](#) (*all boards are fully tested, but they may damage during the shipping, we try to package them as better as possible*), if it works, go ahead with the installation.

Connect the power from your USB charger to the USB-C connector on the board. Then, with a multimeter in voltage measurement mode, **check for a 5V reading**. If that's the case, continue with the installation.

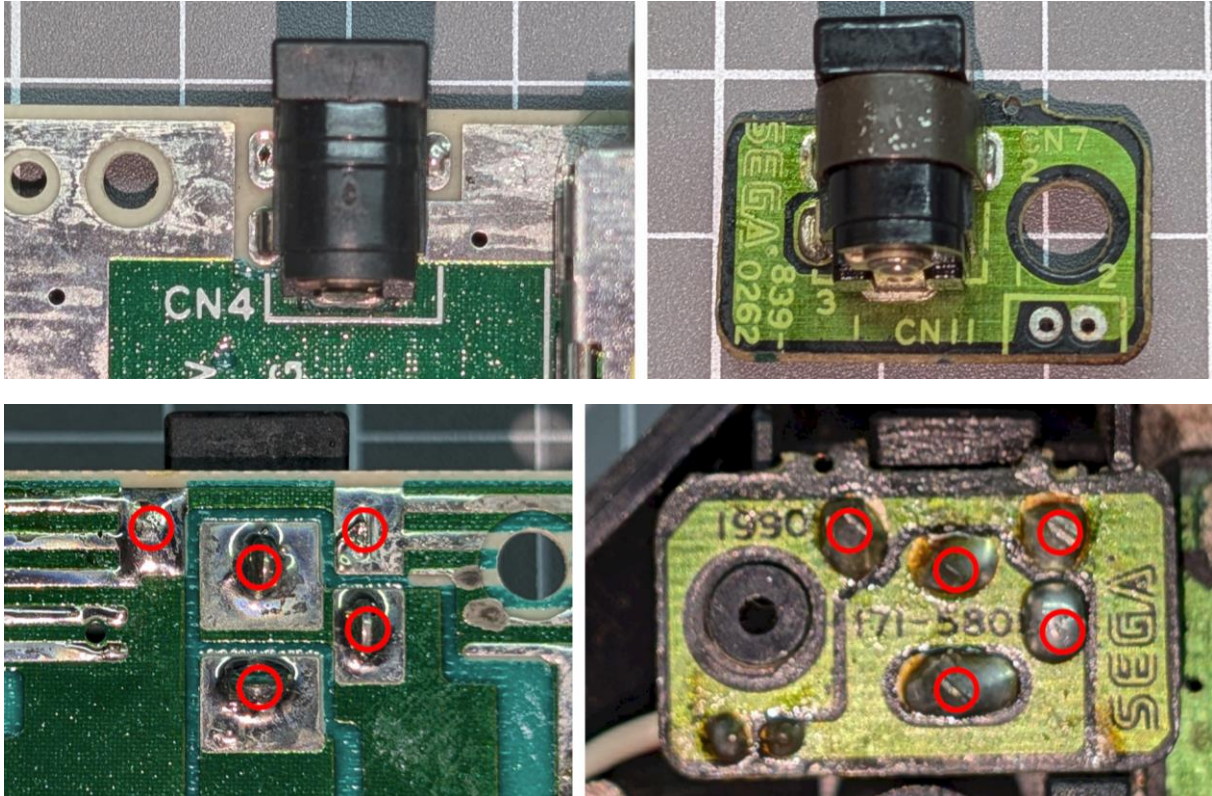


BOARD COMPATIBILITY

There were many mainboard revisions for each of the consoles. It's advisable to check whether the connector and the pin layout match those shown in the following images. This will ensure a successful installation.

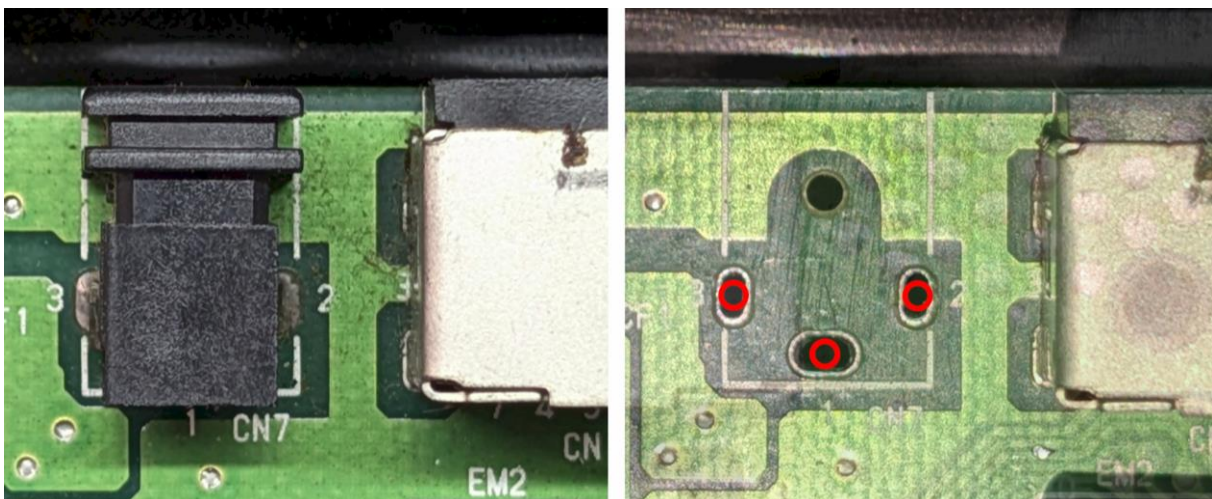
MASTER SYSTEM 1, MASTER SYSTEM 2 & MEGA DRIVE 1

The kit is designed to be used with this connector and connector shape on the mainboard.



MEGA DRIVE 2

The kit is designed to be used with this connector and connector shape on the mainboard.



INSTALLATION STEPS

Please, carefully read the following steps for a successful installation.

PRE INSTALLATION STEPS

Before the installation, your console may need some extra steps to have it ready for the kit.

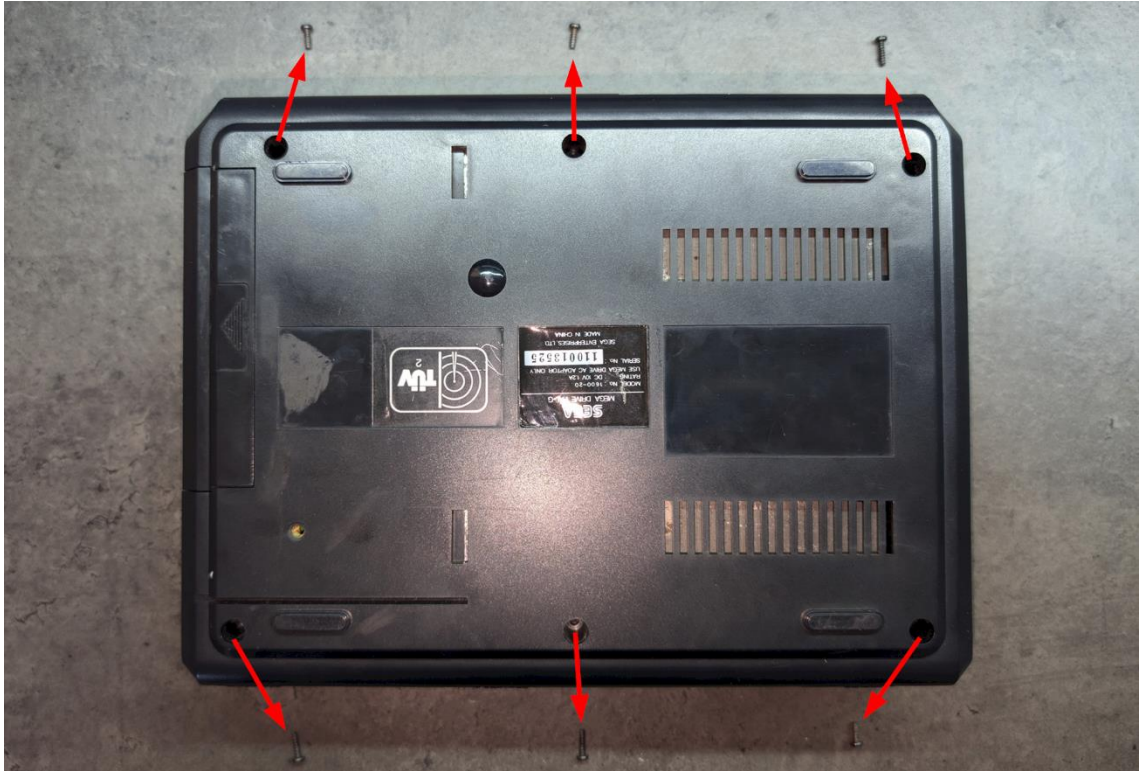
1. DISASSEMBLY THE CONSOLE

The disassembly steps for the Master System 1, Master System 2, Mega Drive 1, and Mega Drive 2 are very similar. We will only show how to do it on one of them.

If you need it, you have the step-by-step disassembly guide on iFixit:

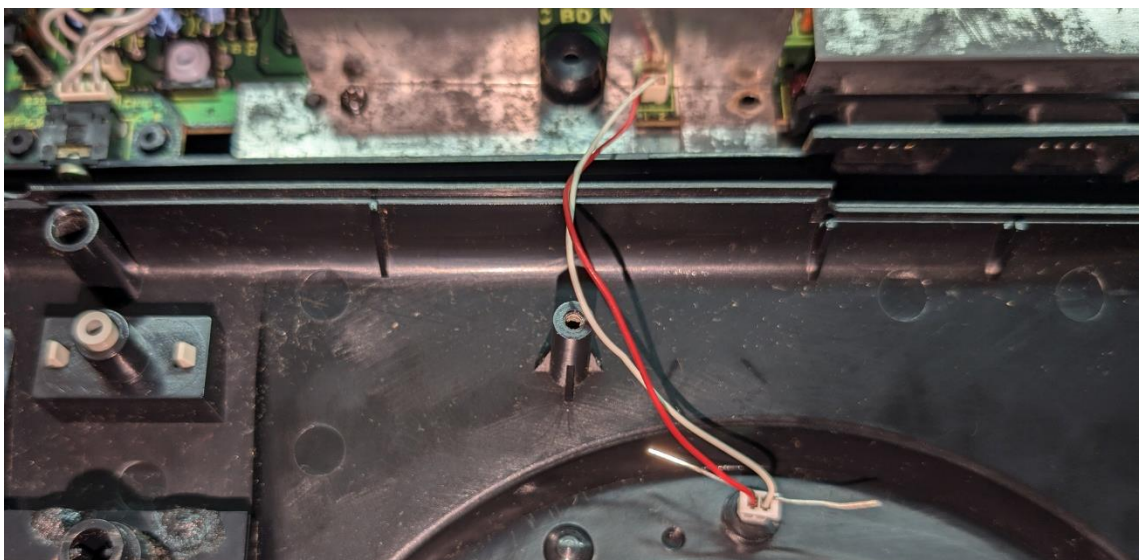
- Master System 1
- [Master System 2](#)
- [Mega Drive 1](#)
- [Mega Drive 2](#)

Remove the Phillips screws from the bottom of the console that hold both halves of the plastic shell together. Depending on the console model, the number of screws may vary. *If you find any screw that is longer or shorter than the others, make a note of its location so you can put it back in the correct place later.*



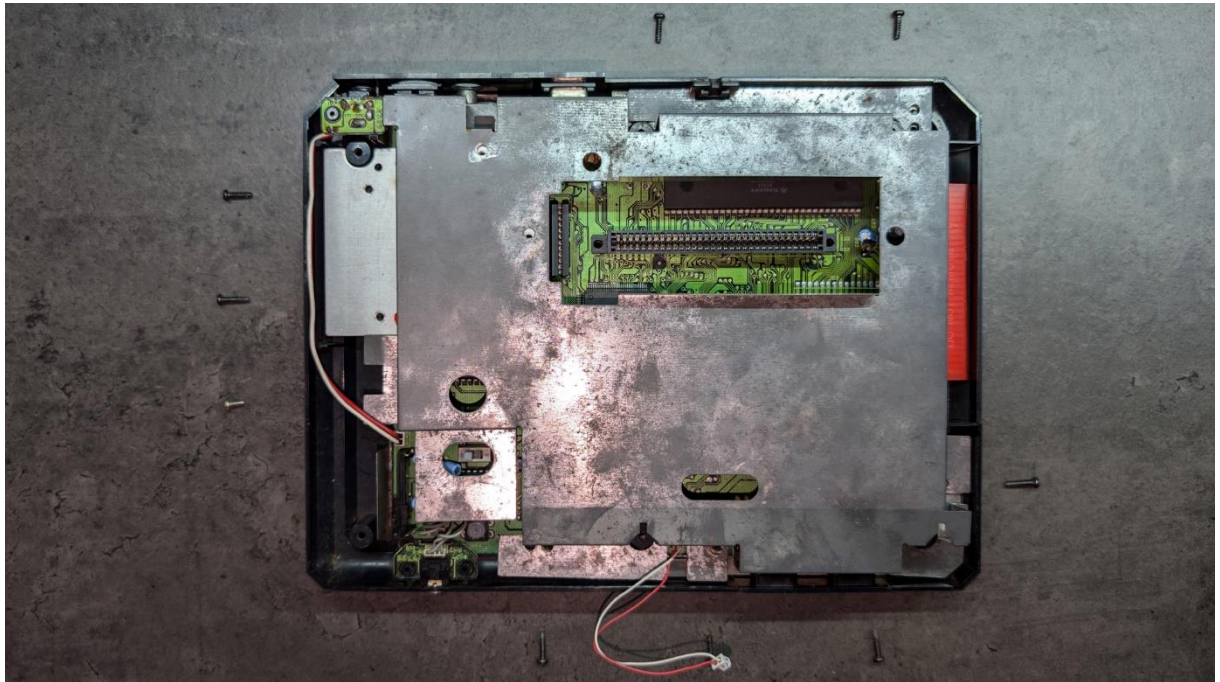
In the case of the Mega Drive 1, be careful when separating the shells, as they are connected by the top LED wire.

To remove the "connector", you'll need to bend the two metal legs of the LED inward to straighten them again. After that, you can pull the connector out. **It's recommended to slightly trim the leg connected to the white cable so you'll know to reconnect it in the same spot later.**



Remove the screws that hold the metal shield and the mainboard to the bottom plastic shell.

Again, some screws are a different length from the others, so make a note of where each one goes.



2. REMOVE UNNECESSARY PARTS

In all four console models, it's essential to remove the original power connector, and also to either remove certain components or at least bypass them so they no longer affect the circuit.

Let's look at each console individually.

POWER CONNECTOR

MASTER SYSTEM 1, MASTER SYSTEM 2, **MEGA DRIVE 1** & MEGA DRIVE 2

The power connector is removed the same way on all consoles, **except for certain specific versions of the Mega Drive 1 mainboard. If your MD1 has the connector separated from the mainboard, then skip to the next section.**

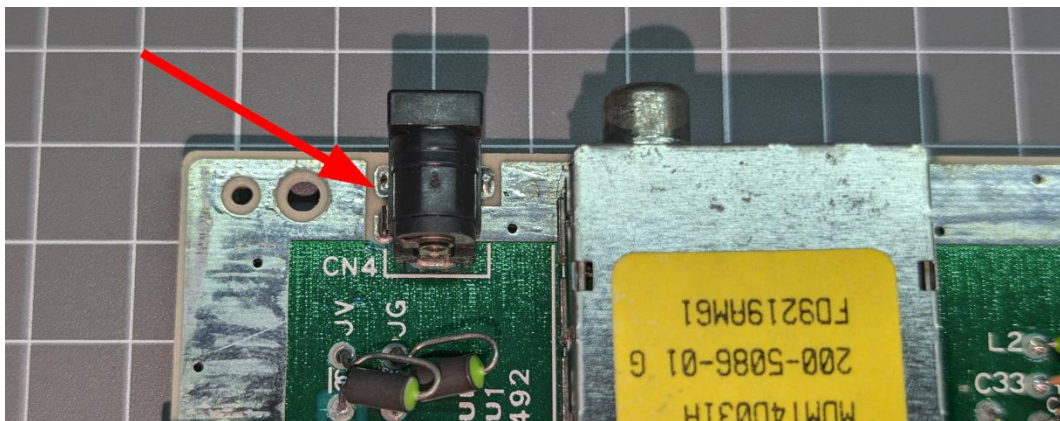


Image of the Master System 2

Apply plenty of solder to all the pins of the connector, this way, the new solder will melt the old one and also allow you to heat all the pins at the same time. While doing this, pull the connector from underneath using tweezers or pliers to remove the old connector. If it comes apart or melts due to the heat, don't worry, it won't be used again.

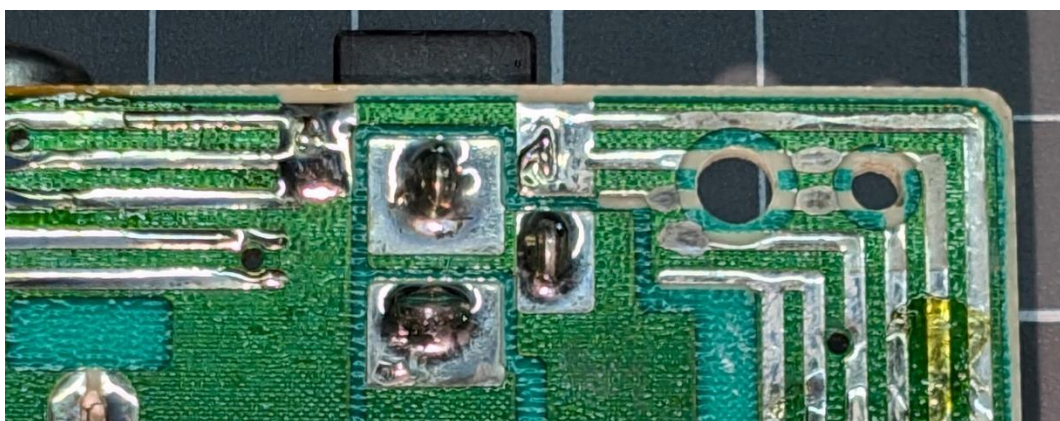


Image of the Master System 2

Once the connector has been removed, make sure all the holes are free of solder and that the top surface is flat, that is, there should be no bumps or leftover solder that make the surface uneven.

Clean both sides with alcohol.

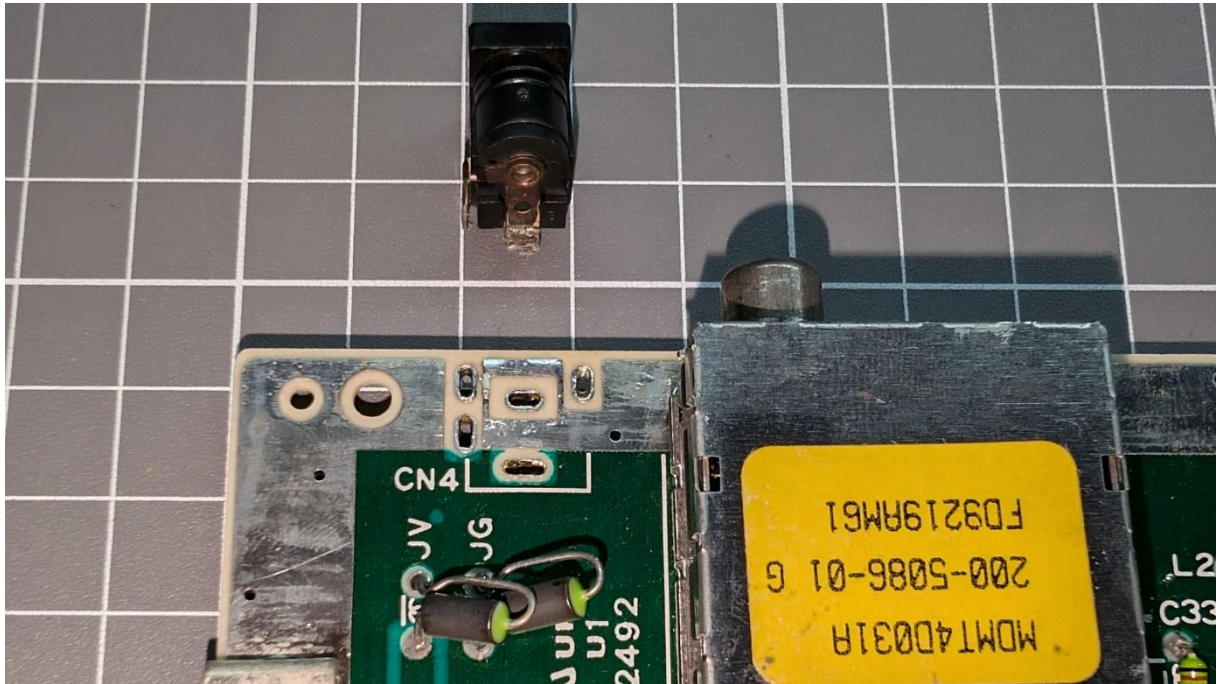


Image of the Master System 2

MEGA DRIVE 1

In this case, it's even easier; you'll see that the connector is on a separate board from the mainboard. You just need to desolder the connector, as we will replace the entire mini board with the new one included in the kit.



Image of the Mega Drive 1

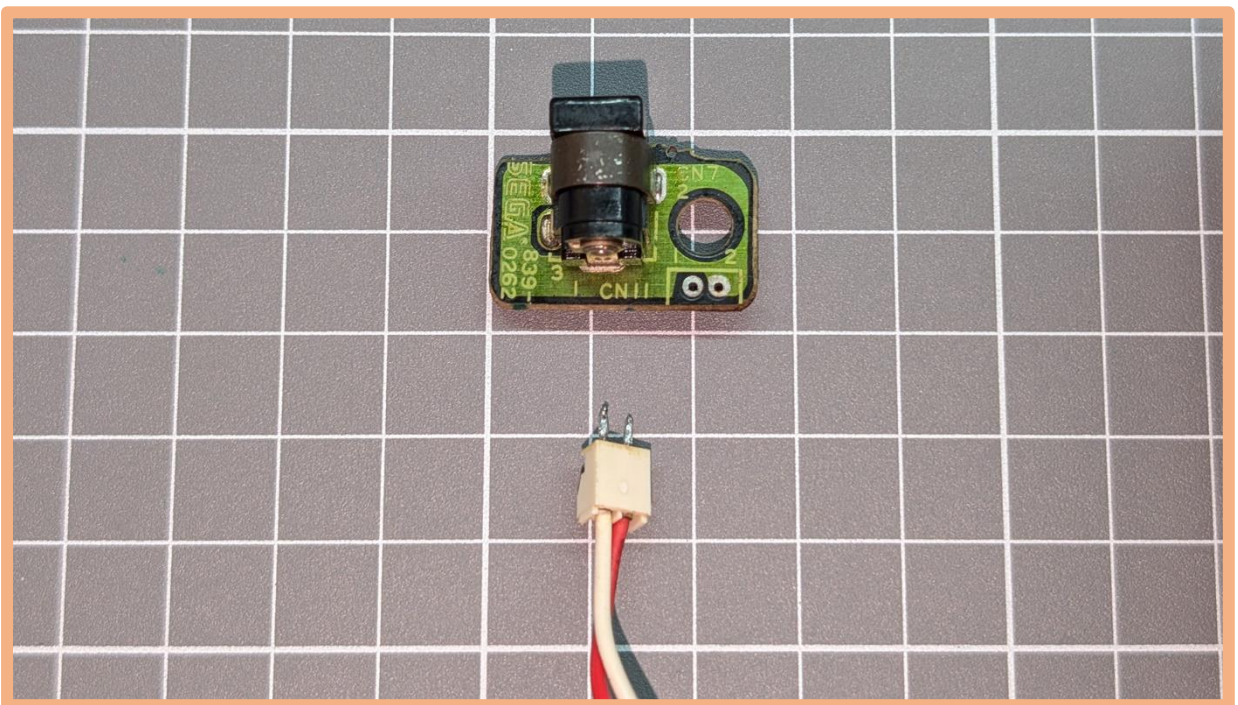


Image of the Mega Drive 1

OTHER COMPONENTS TO REMOVE

In addition to the power connector, it is necessary to disable or remove some components from the mainboard that cause the input power to decrease, making it insufficient for the proper operation of the console through USB-C. These components were required when the power was higher with the original power supply, but now that the console will be powered by 5V from USB-C, they are no longer needed.

The easiest way is to use the cables included in the kit to bypass the input and output pins of each component; just by doing this, the component will be disabled and will no longer perform any function.

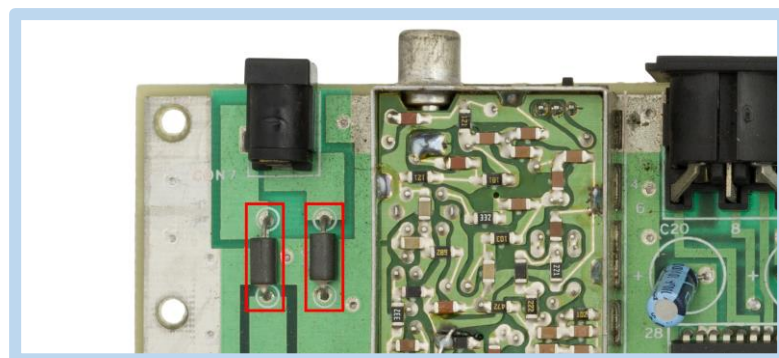
If you prefer, you can remove the component from the mainboard and then solder the wire using the holes left by the removed component. However, this is very difficult to do for voltage regulators, as a lot of heat is required.

MASTER SYSTEM I

Unfortunately, I didn't manage to get this console in time. I will update the manual with photos once I get it.

The components to bypass are the same as those in the Master System 2.

First, these two components along with the power connector. Bypass each side of the component on the bottom side of the mainboard.



And also, the voltage regulator. **Bypass the I and O pins** by connecting them together on the bottom side of the mainboard.

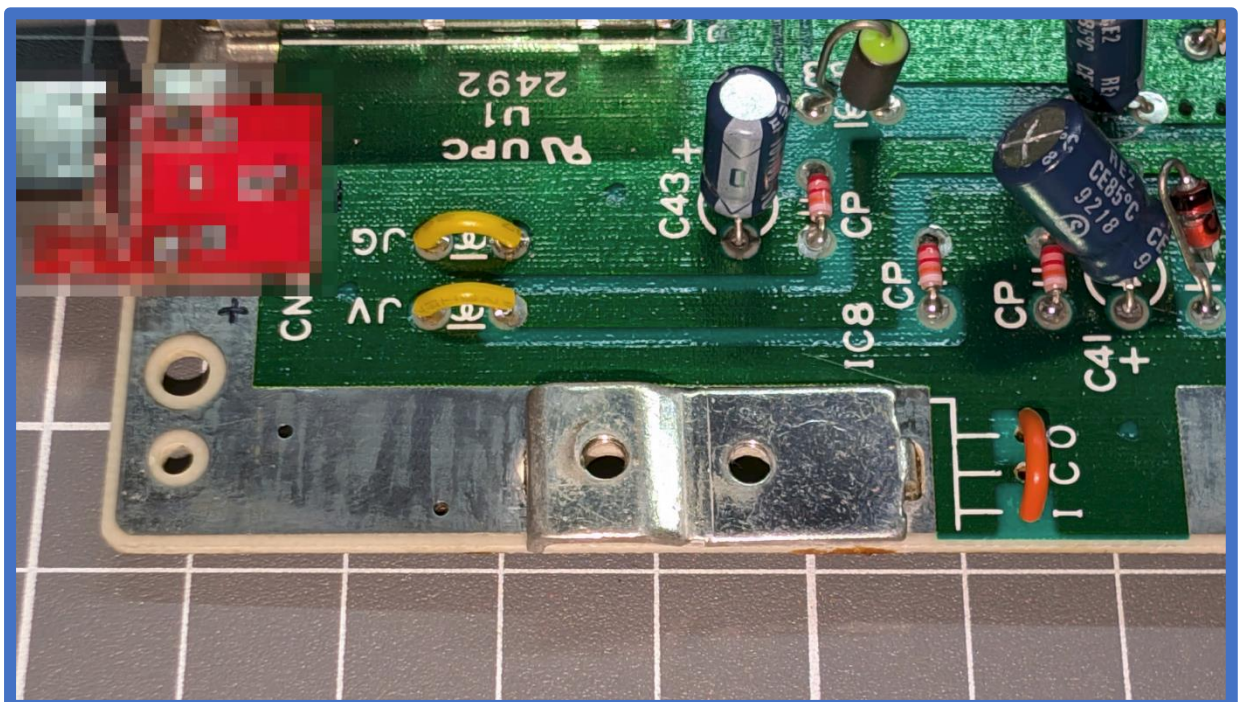
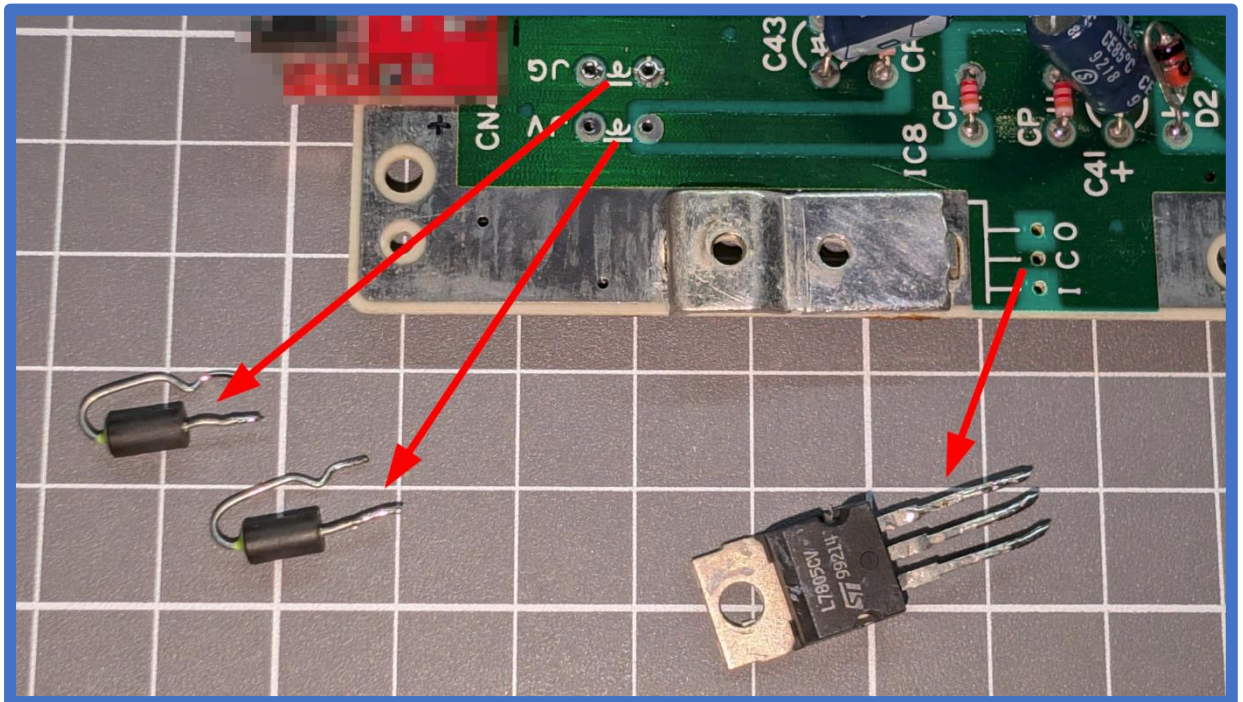


MASTER SYSTEM 2

Remember, you can simply bypass the components or, as I did, remove the component and then solder the wire for a cleaner and more labor-intensive installation.

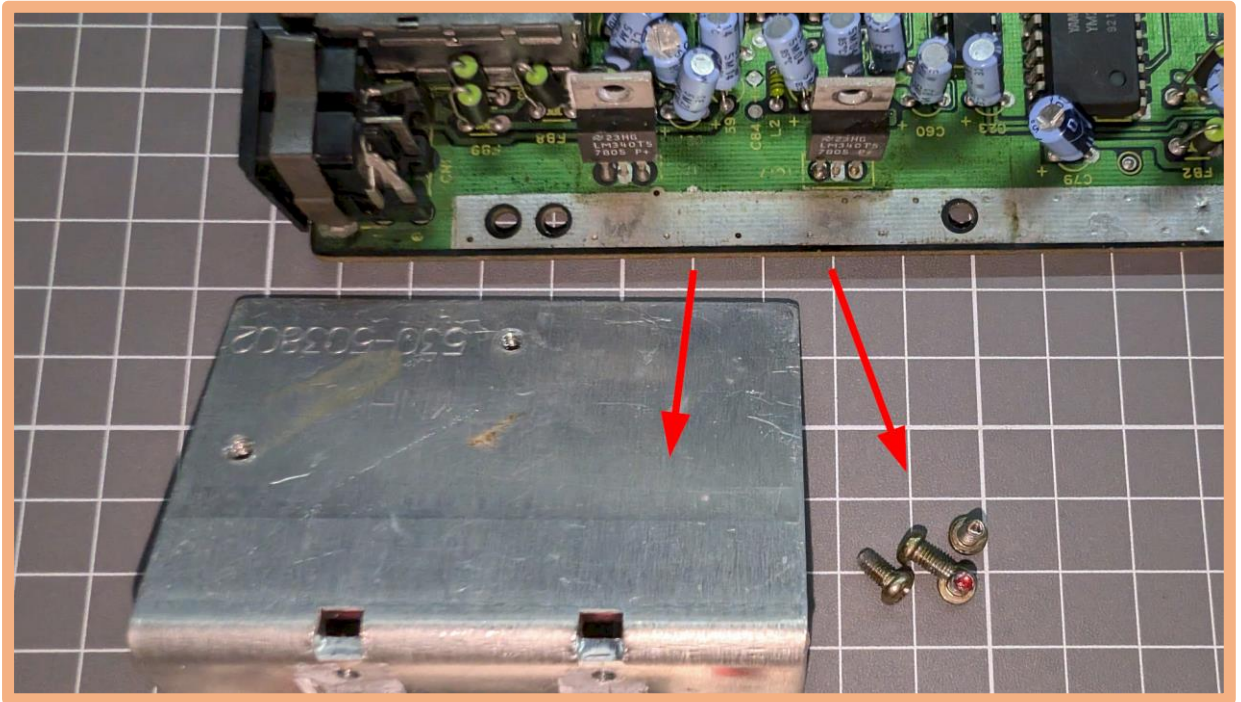
Bypass or remove the two components next to the power connector, as well as the voltage regulator, where you'll need to bypass the I and O pins.

Ignore the fact that the USB-C board is already installed; that will be explained later :)

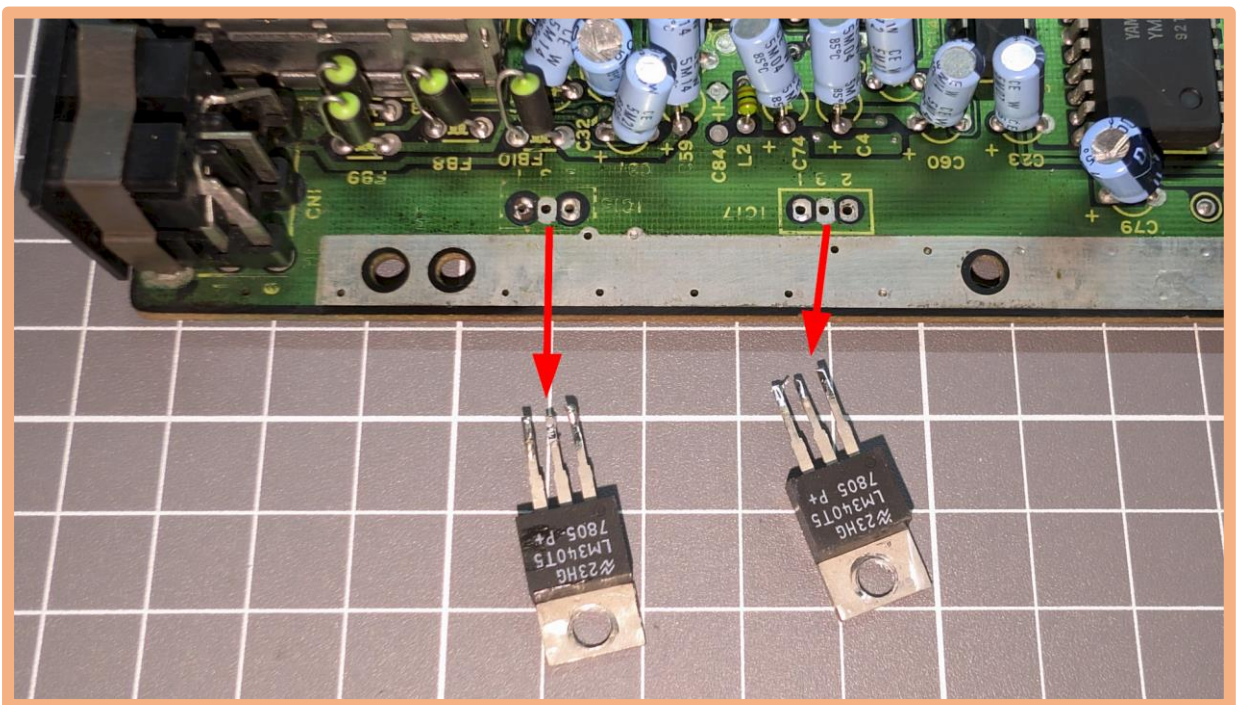


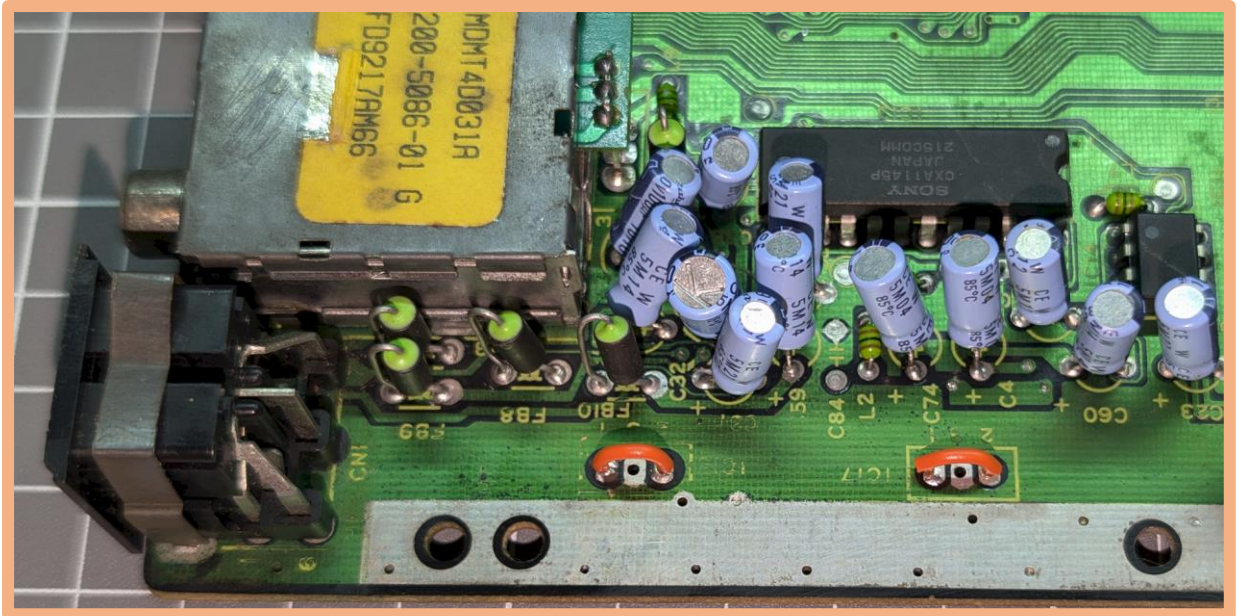
MEGA DRIVE I

You can bypass the components or remove them. If you choose to remove them, you can also take off the heavy heatsink, as it's no longer needed.

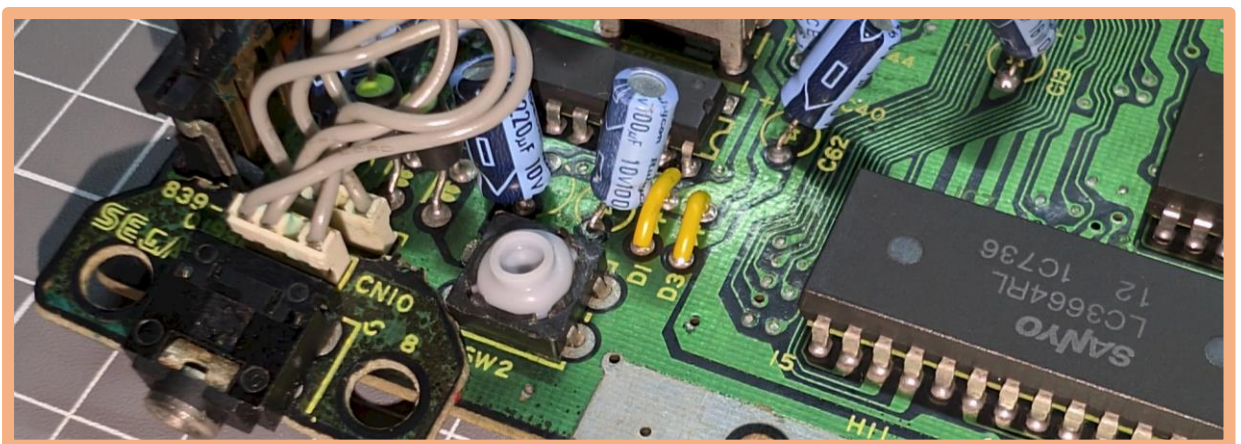
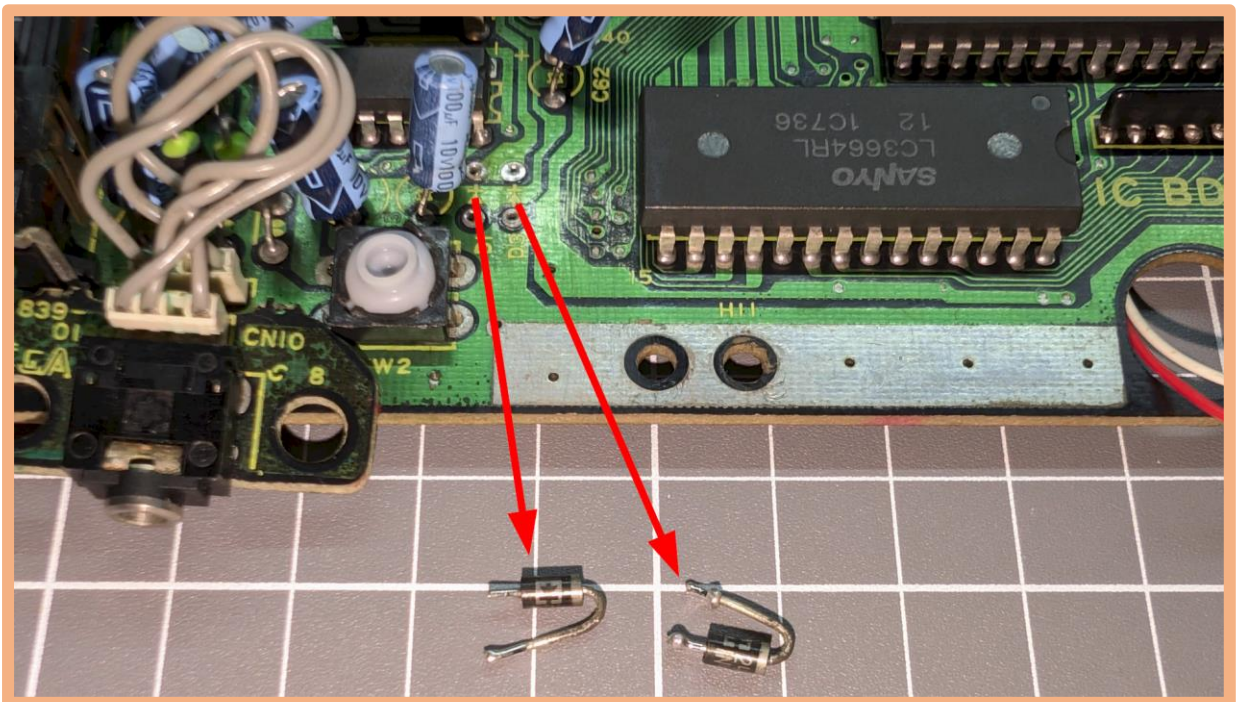


In this case, you'll see there are two voltage regulators, this was simply because a single one couldn't provide the current the console required. You will need to bypass both.



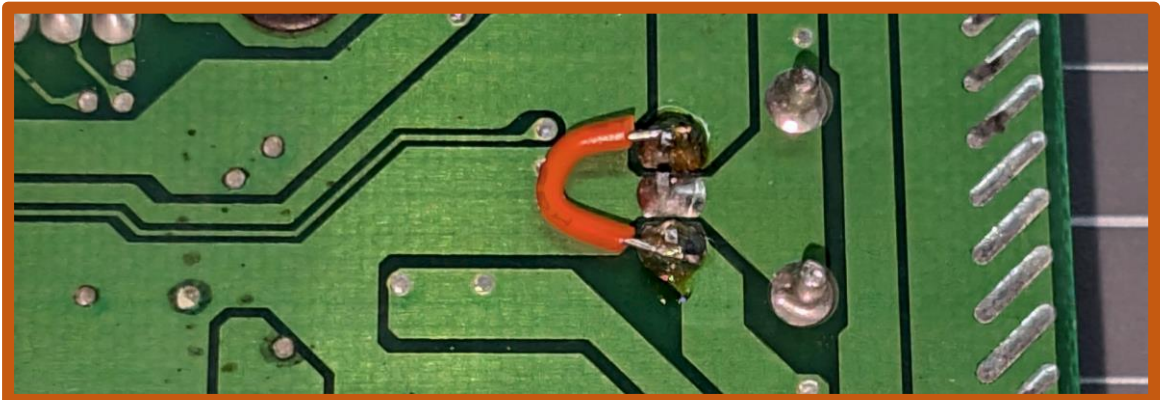
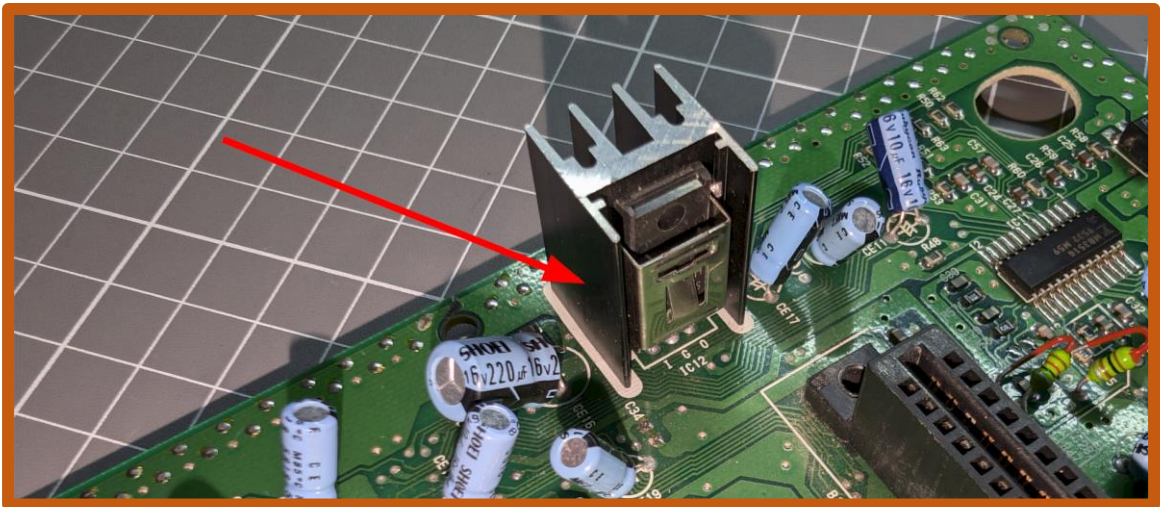


These two diodes to bypass or remove are next to the reset button.

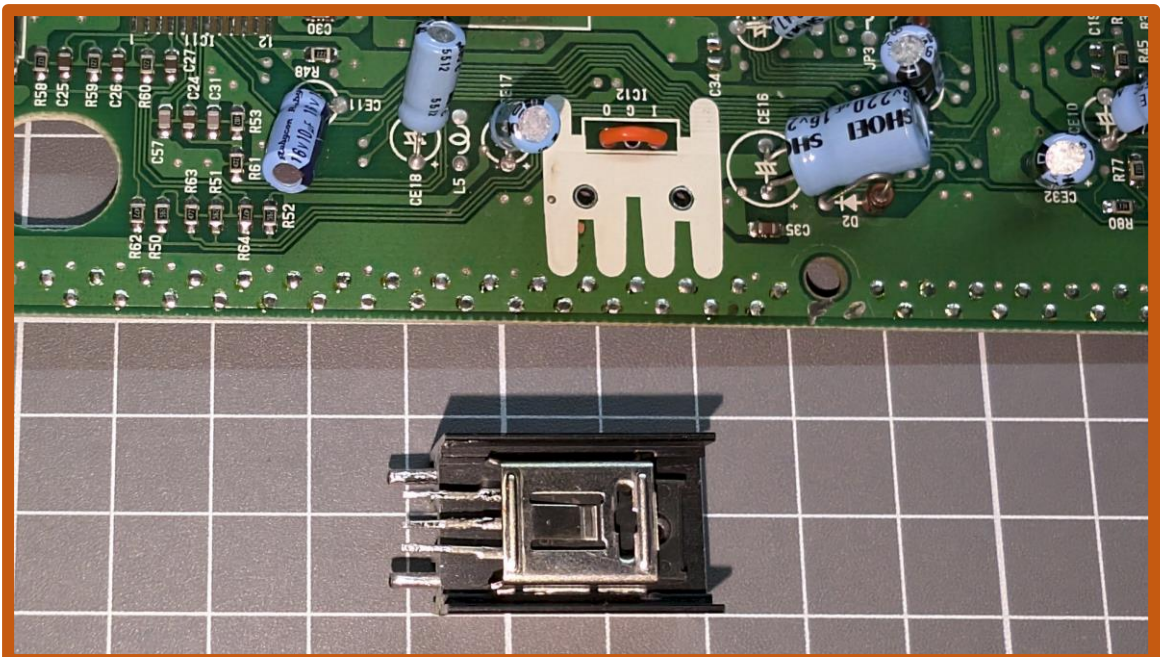


MEGA DRIVE 2

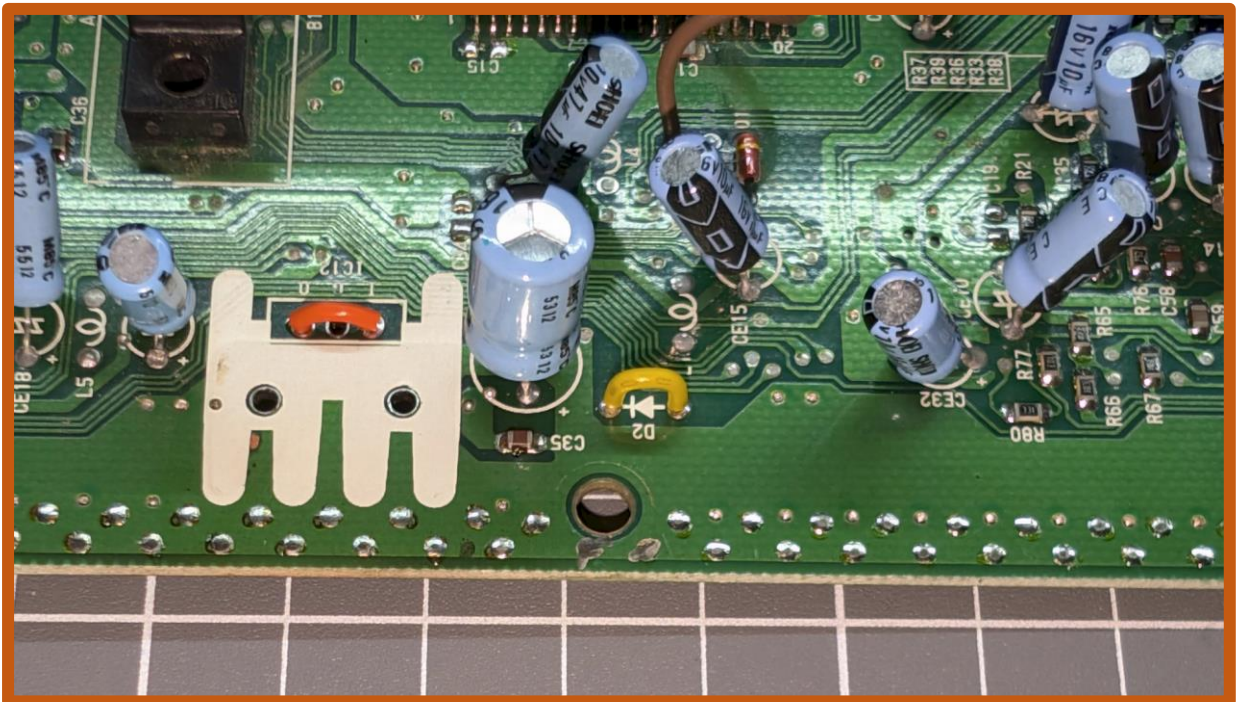
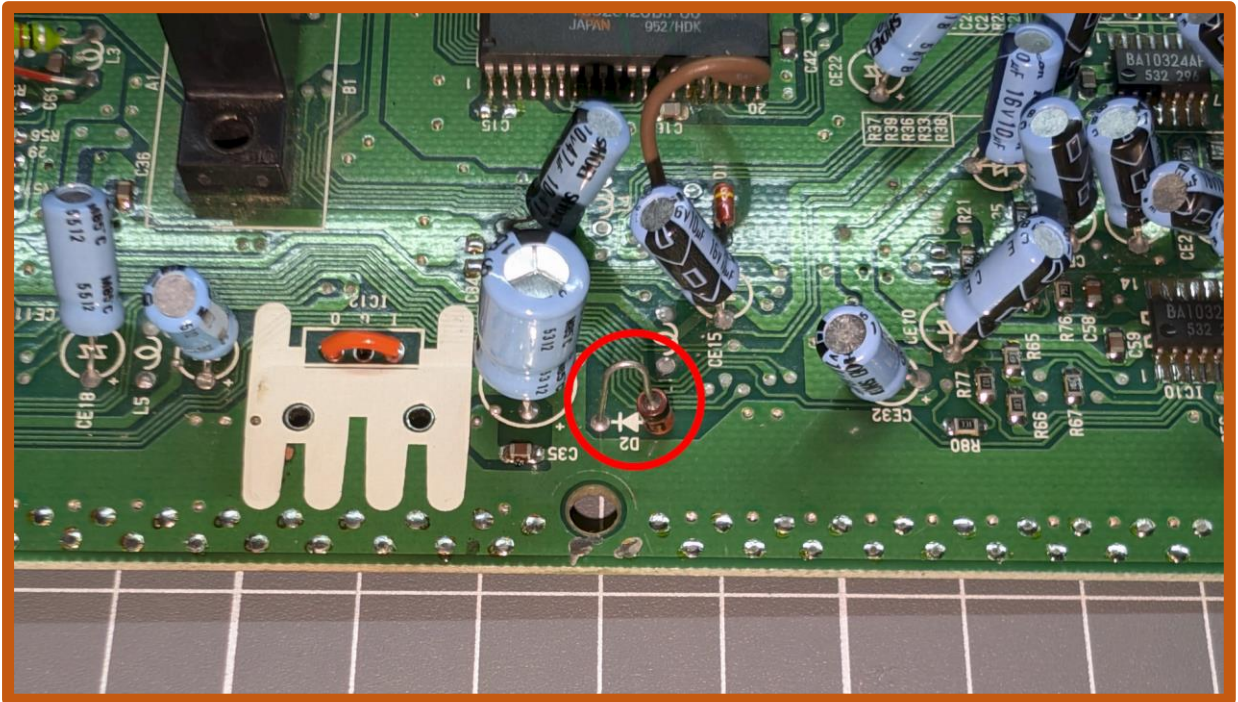
Bypass or remove the unnecessary components. In this case, I recommend that the voltage regulator only be bypassed, and not removed, as it is very difficult to take out.



That said, using a hot air rework station and a soldering iron at the same time, it can be removed.



This component also needs to be bypassed or removed.



INSTALLATION STEPS

1. BOARD ASSEMBLY

Now it's time to finish assembling the USB-C board.

The USB-C board has many pads; pins should only be soldered to the pads needed for the installation.

MASTER SYSTEM 1, MASTER SYSTEM 2 & MEGA DRIVE 1

Solder 4 pins only to the pads with a **round shape**.

MEGA DRIVE 2

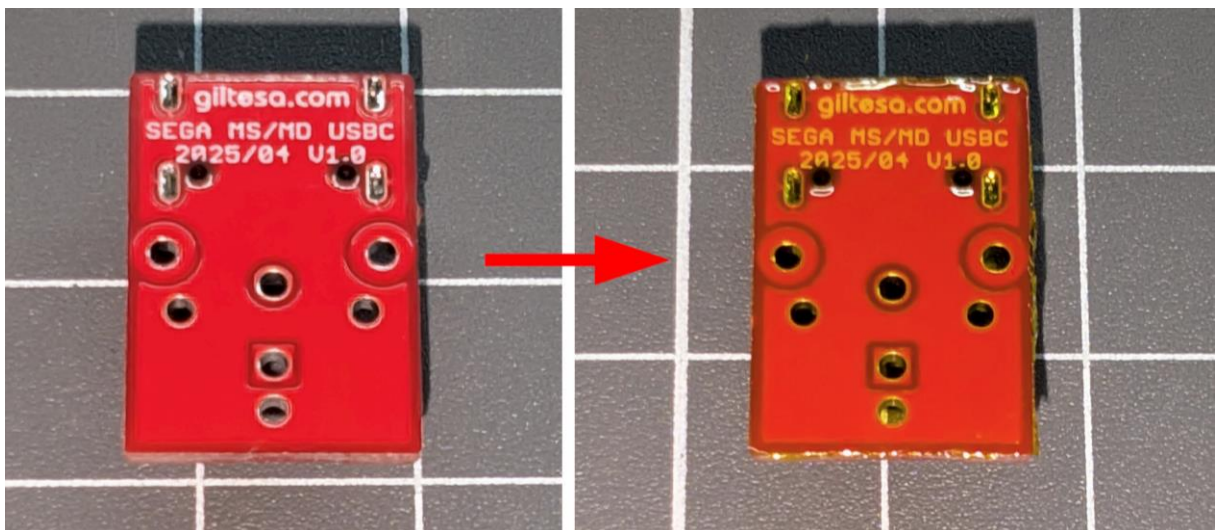
Solder 3 pins only to the pads with a **square shape**.

The way the pins are soldered doesn't change, so it's explained only once. In this case, solder them where appropriate.

ISOLATE THE CIRCUIT WITH KAPTON TAPE

In addition to isolating both circuits to prevent accidentally bridging adjacent pads, the Kapton tape will also help keep the pins from moving too much, making them easier to solder later.

Cover the bottom side of the circuit with Kapton tape. Trim the excess tape so it doesn't stick out.



Finally, use a cutter or a punch to make a hole in the pads where the pins will be soldered later.

SOLDER PINS

Insert the pins from the bottom side, leaving the plastic on that side, just as shown in these two images:

MASTER SYSTEM1/2 & MEGA DRIVE1



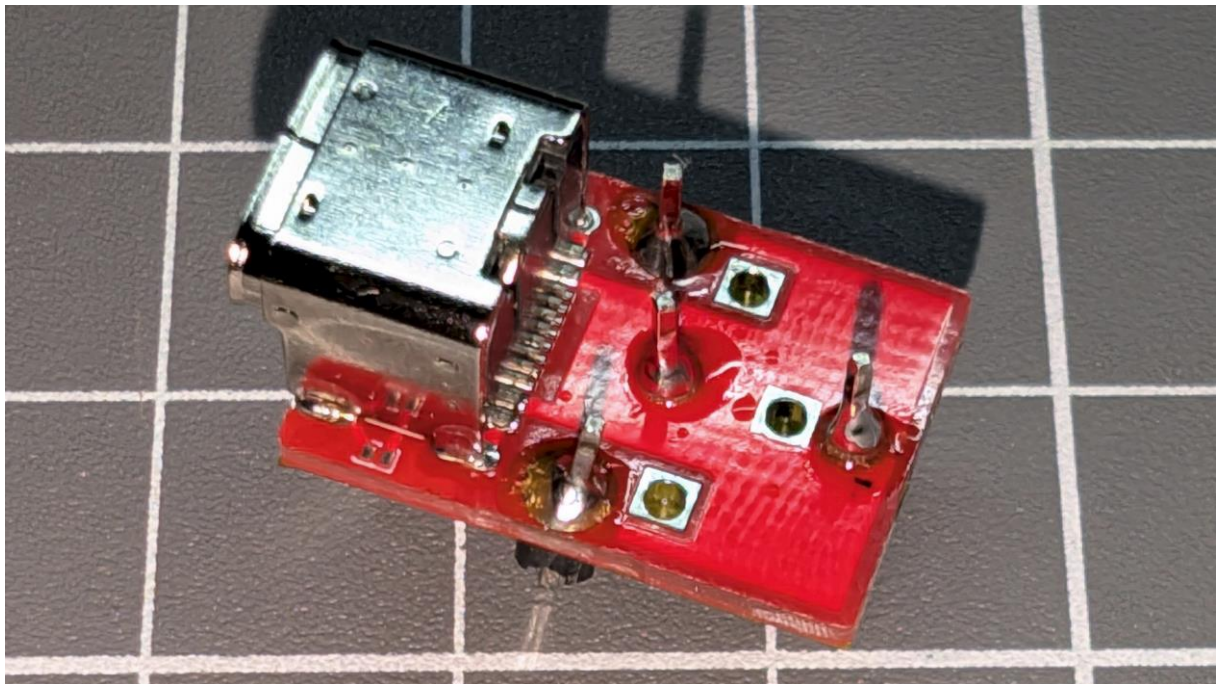
Use only the round shape pads!

MEGA DRIVE 2

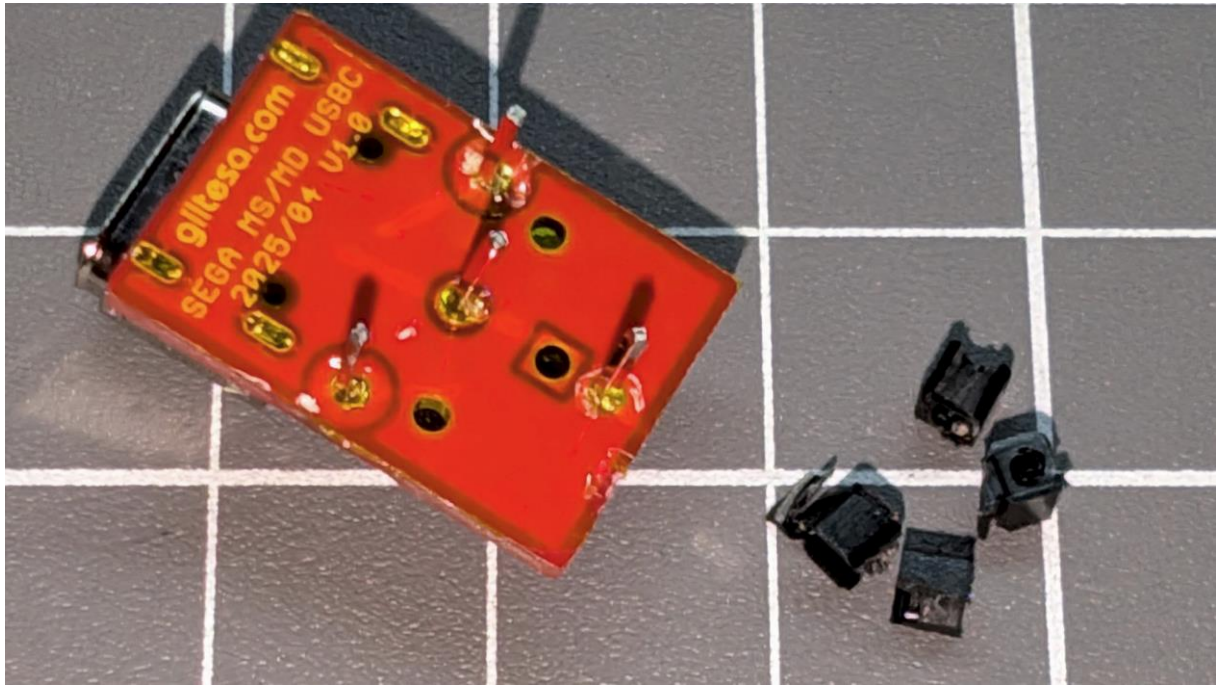


Use only the square shape pads!

Once the pins are placed, flip the board and solder each one. Make sure they are straight/upright. The pin length on each side doesn't matter too much, as it will be trimmed later.



Now, remove the plastic from each pin.

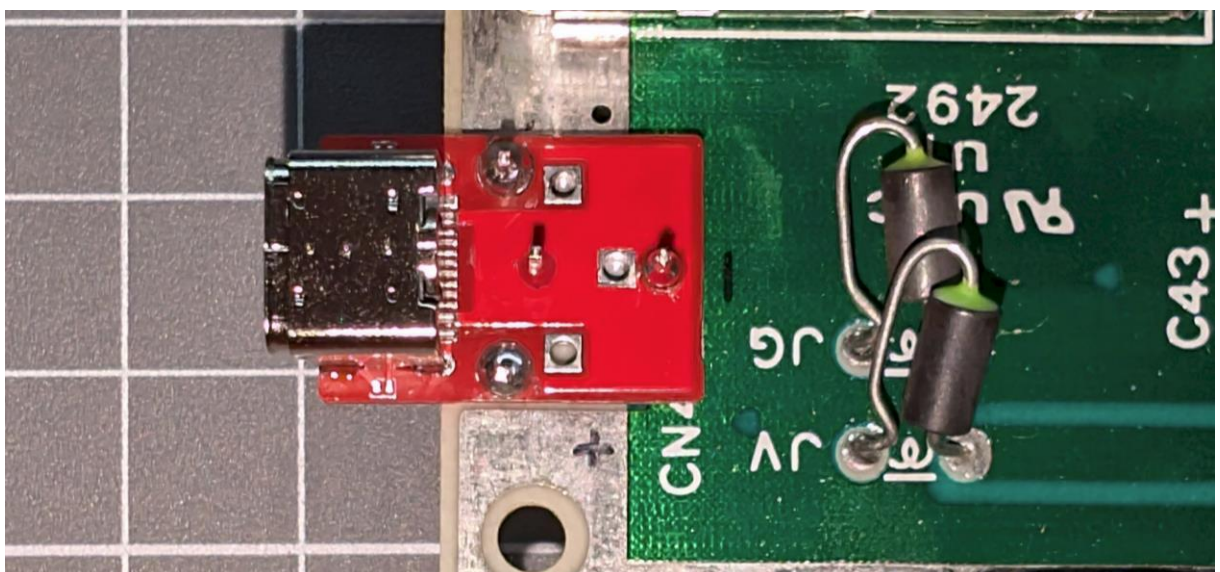


SOLDER THE BOARD TO THE MAINBOARD / OR TO THE MEGA DRIVE 1 ADAPTER

Now it's time to solder the USB-C board to the console's mainboard, or if your console is a Mega Drive 1 with a separate power board, then solder the USB-C board to the adapter included in the kit.

The soldering method is the same whether you are soldering to the mainboard or the MD1 adapter.

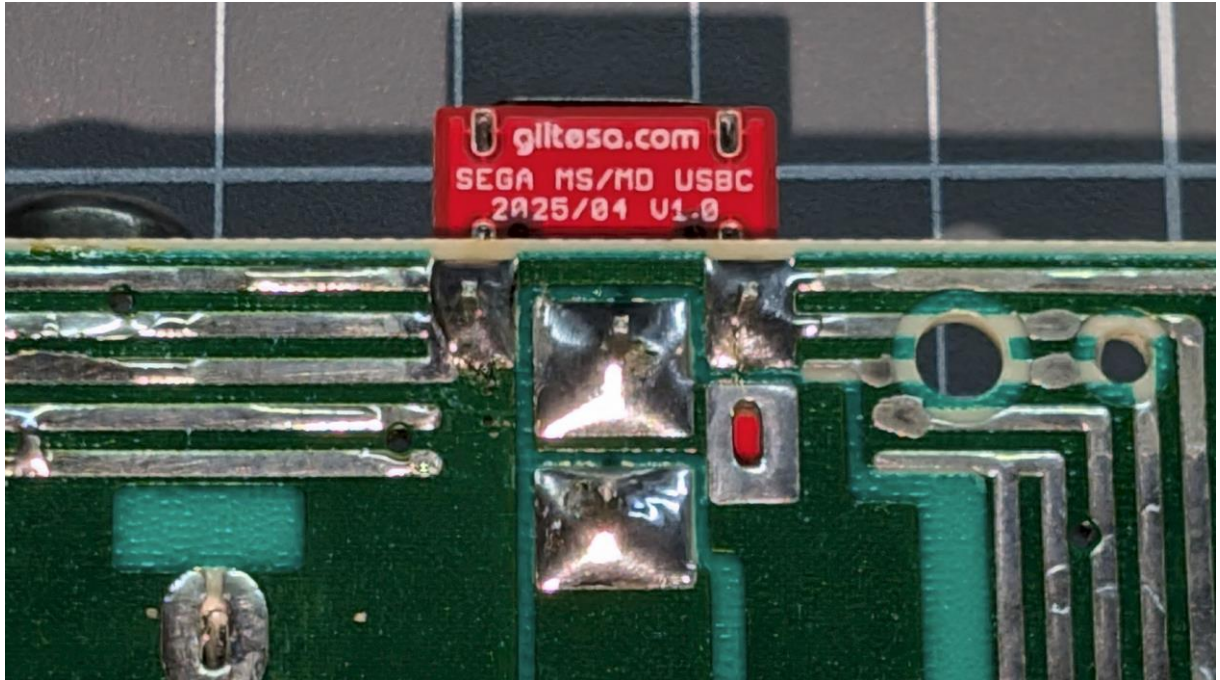
Place the board on top of the other board, ensuring that the pins align with the holes and that both boards are in direct contact and completely flat with each other.



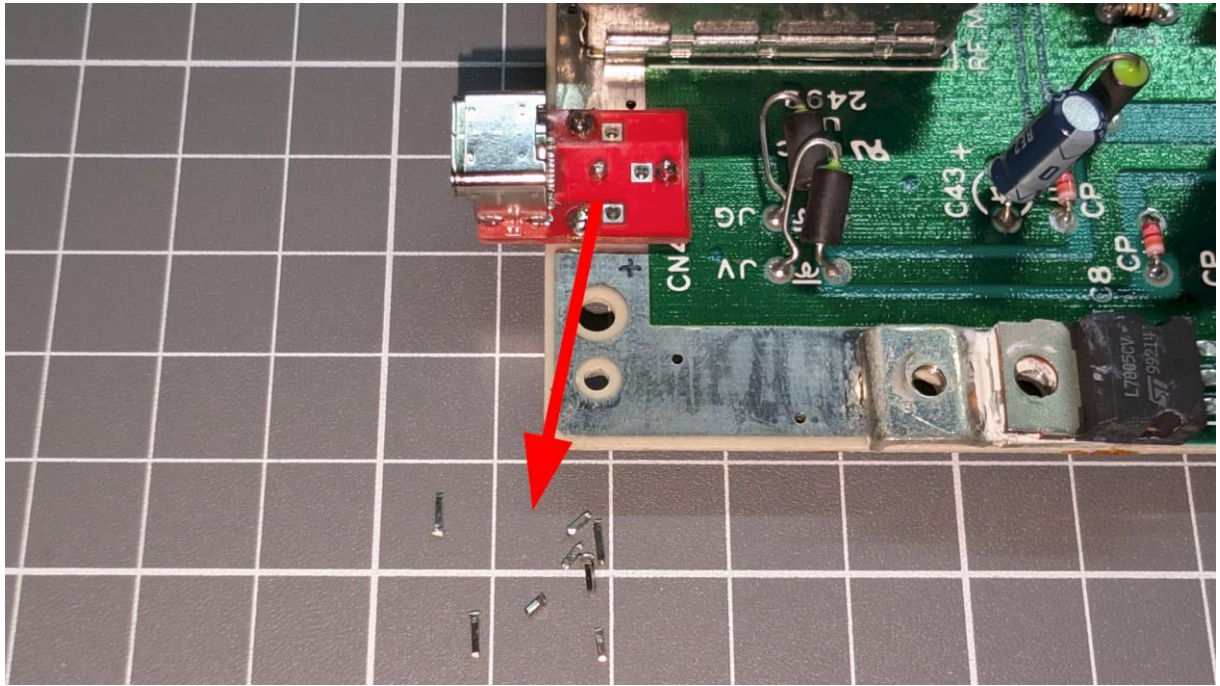
Solder only one pin. Verify that the board is aligned and parallel to the mainboard. As you can see in this image, both boards are not parallel yet and will be corrected later.



Once corrected, solder all the pins, always ensuring that both boards remain flat against each other.



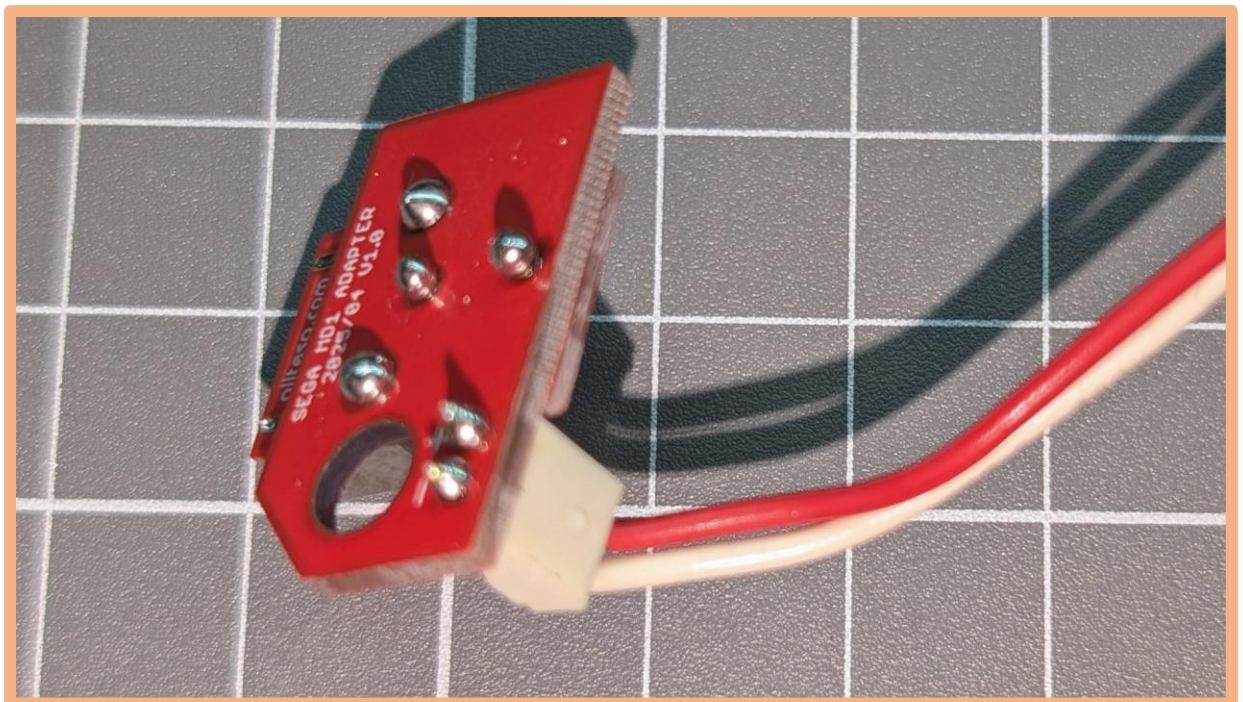
Finally, trim the excess of the pins on both sides.



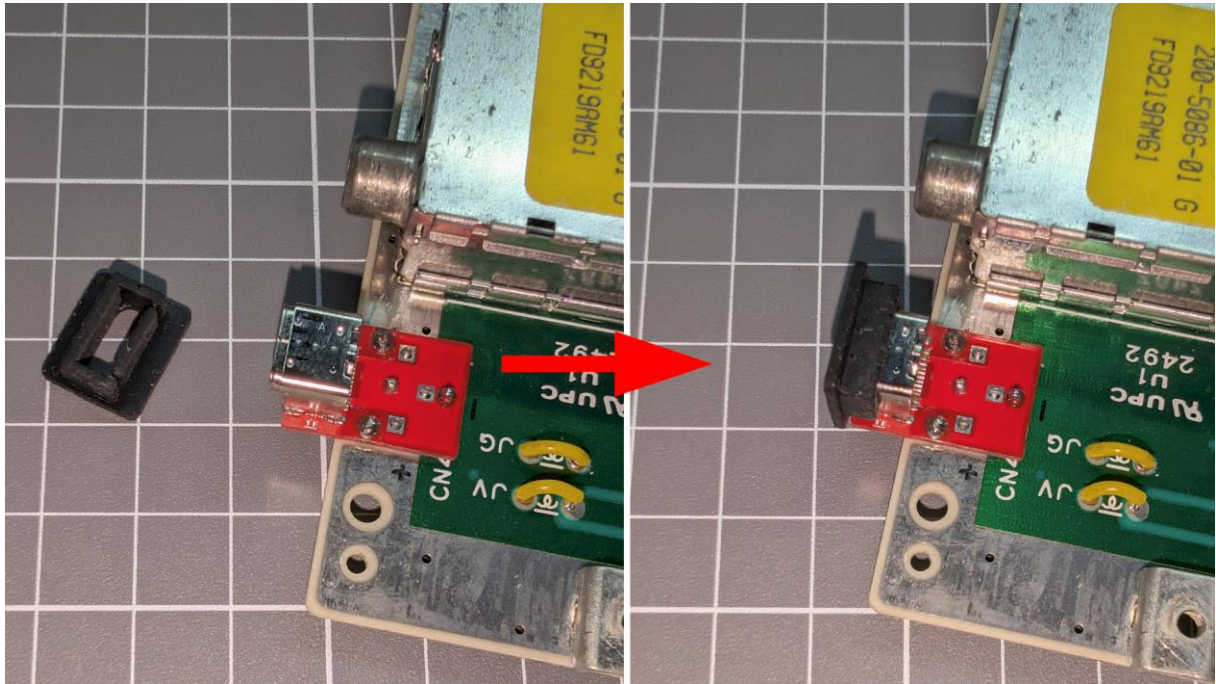
On the Mega Drive 1 with a power connector separated from the mainboard, you'll need to solder the USB-C board to the adapter included in the kit. It is soldered following the same steps used to solder the USB-C board directly to the mainboard, only this time it's done on the adapter.



Then solder the console's power cable to the adapter board, respecting the polarity. The **red cable goes to the pad marked with a +**, and the **white one to the pad marked with a -**

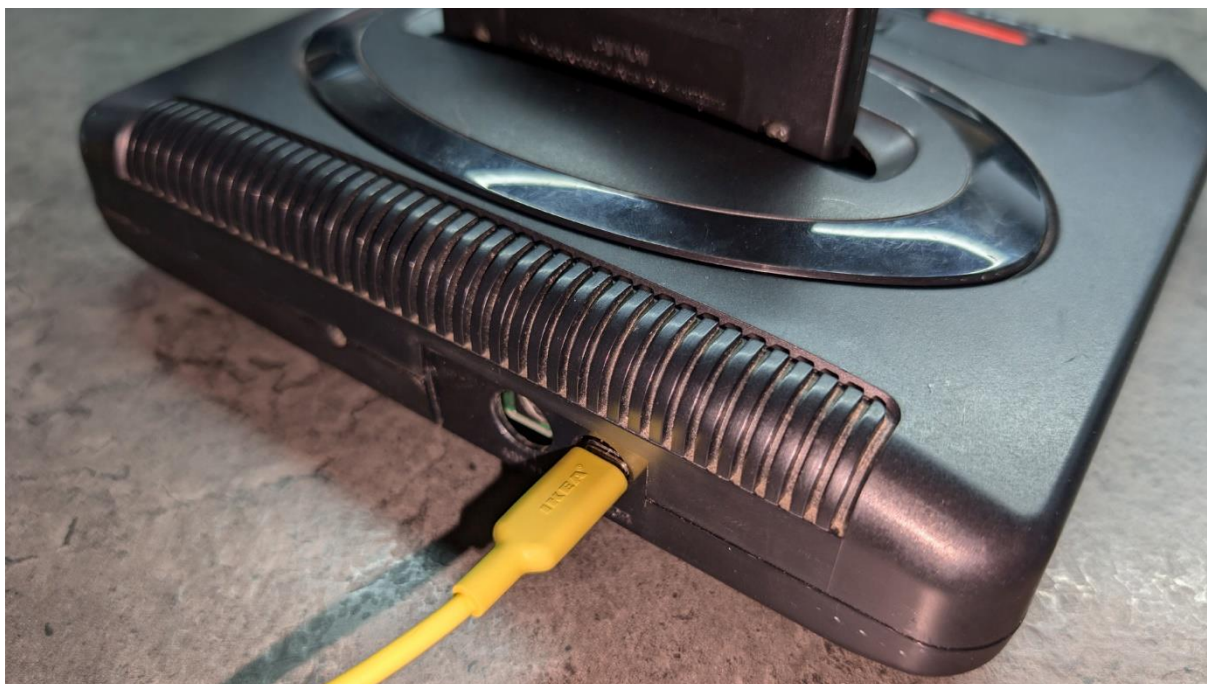


Don't forget to place the plastic cap on the USB-C to cover the hole.



2. **DONE!**

The installation is complete. Reassemble the mainboard inside the casing, place each piece where it was, secure everything with the screws, and enjoy your console powered by USB-C.



FREQUENTLY ASKED QUESTIONS - FAQ

WHAT CHARGER CAN BE USED?

You can use any standard charger for mobile phones, computers, etc., with 5V 1A. It doesn't need to be a Power Delivery charger since this feature is not used. Of course, if you want to use a Power Delivery charger, there's no problem or risk.

Technical data for curious minds:

Power Delivery chargers can supply a wide range of voltages: 5V, 9V, 12V, 15V, and 20V. However, for this to happen, the device must communicate with the charger to explicitly request the desired voltage. Without this communication, the charger will never supply more than 5V. That's one of the advantages of USB-C, as it can be used with both old and modern devices.